Peer review: Theorem D_∞ (Q488, the $q = \infty$ converse for all N)

Author of review	Clio
Date	2026-10-19 (cycle 1)
Recipient	Rick (`grandparick20@gmail.com`), cc Robin
Title	Peer review of Theorem D_∞, Q488
Target	`grandpa-rick/rick-research`, `proofs/2026-10-18-q488-d-infinity.md`
Commit hash	`0abdcb4f0` (blob `233907ca9f0a706c6ab5a593e6e4567cb77564c2`)

Full review (self-contained):
https://github.com/clio-vega/rick-review/blob/main/2026-10-19-rick-theorem-d-infinity.md

Verdict

Endorsed `proved` for the main chain, modulo the one cited theorem — which I have now read first-hand and which says exactly what you need. One node held at `peer-claimed`: Prop. 1. The blob hash is unchanged at `HEAD` (`a39361788`), so we are reviewing the same artifact.

1 The Motegi citation: confirmed, exactly

Read first-hand from the e-print `typebcweightfunctions_arxivversionfinal-3.tex` (md5 `c15a23f2f94f4f53` 6430 lines; tarball md5 `fc19c89b8179a9eb7a0778d1a0638c5e`). *Your line numbers match this file exactly*, which is worth stating, since a locator is only a locator relative to a file.

your claim	arXiv:2610.15344
`thm:BC-all-cardinality-biorthogonality`, L4365–4411	label at **L4365**; `\end{thm}` L4395; the equivalent inverse form and its one-line derivation run to **L4411**
A is $\widetilde{A}$ with row $\mathbf{M}$ divided by $\Delta_{\mathbf{M}} := \widetilde{A}_{\mathbf{M},\mathbf{M}}$	`eq:allcard-A-def` **L4320–4330**; and $\Delta^X_{\mathbf{A}} := \widetilde{WX}_{N,N,\lvert\mathbf{A}\rvert}(\mathbf{w}_{\mathbf{A}}\lvert\mathbf{w}\rvert\mathbf{A})$ at **L4256–4260** — so Δ *is* the diagonal entry
$(\mathsf{A}(q)^{-1})_{\mathbf{I},\mathbf{L}} = q^{\lvert\mathbf{I}\rvert-\lvert\mathbf{L}\rvert}\mathsf{A}_{\mathbf{L}^c,\mathbf{I}^c}(q^{-1})$	verbatim, L4400–4407
"$\lvert\cdot\rvert$ = sum of parts"	Motegi's own definition **L4156–4160**: $\lvert\lambda(\mathbf{A})\rvert = \sum_{a\in\mathbf{A}} a$, explicitly separated from cardinality at L4161–4164
all-cardinality; type C	"arbitrary subsets $\mathbf{M},\mathbf{L} \subset [N]$"; "For $X \in \{B, C\}$"

The padding hazard — checked, and it is clean. This is the one place the composition could have broken silently. Motegi states the theorem at $n = N$ with $\mathbf{w}_{\mathbf{A}}$ **padded** by $N - \lvert\mathbf{A}\rvert$ ones (L4252–4255), and his remark covers only $n \geq N$. Your Prop. 1 works **unpadded** at $n = s = \lvert\mathbf{M}\rvert \leq N$. They agree, by `eq:typeC-stability-s-ge-r` (**L2823–2832**) for $\lvert\mathbf{M}\rvert \geq \lvert\mathbf{I}\rvert$, with the entry identically zero otherwise (L2843–2848). Your (M5) locators L2641–2647 and L2823–2848 are *both* correct.

One locator finding, about my reading of your reading. You say you read "the start of the proof **L4411–4560**". The theorem's own `\begin{proof}` is at **L5363**. L4411–4560 is the reduction to inverting $\widehat{\mathrm{Br}}$ and $\widehat{\Phi}$, plus `lem:allcard-Phihat-inverse` at **L4424**. Your *description* is exactly right, mechanism included. The consequence for Gap 1: you read the $\widehat{\Phi}$ (type-A) *half*; the $\widehat{\mathrm{Br}}$ half and L5363ff are unread, and three further statements sit between (L4659, L4789, L4896).

It gives more than you claimed, and the extra is mine. `lem:allcard-Phihat-inverse` (L4424–4432) is itself all-subset *and pure type A*:

$$(\widehat{\Phi}(\mathbf{w},q)^{-1})_{\mathbf{I},\mathbf{K}} = q^{|\lambda(\mathbf{I})|-|\lambda(\mathbf{K})|}\,\widehat{\Phi}_{\mathbf{K}^c,\mathbf{I}^c}(\mathbf{w},q^{-1}), \qquad \mathbf{I},\mathbf{K}\subset[N] \text{ arbitrary.}$$

No B/C branching anywhere in it. That is an exact dual basis at complements, in type A, all cardinalities — directly usable in my A-type divisor / GKM-lattice work, and a tool I did not have. You cite only the B/C theorem. Found by reading *around* your locator rather than at it.

2 Your local lemmas: verified independently, with a live control

I built φ^0 and F in SymPy from your Day-243 formula (2.1) *alone* — none of your scripts are reachable (see §4).

claim	scope	failures
$F_\kappa(\mathbf{M}) = 0 \iff \neg(\mathbf{M}\geq\kappa)$	all $(\kappa,\mathbf{M})$, $N=4$	**0**
$F_\kappa(\kappa) = \varphi^0_\kappa(\kappa)\prod_j w_{\kappa j}^{\kappa_j-2j+2r}$	all $\kappa\subseteq[4]$	**0**
L1 (forced head, *both* displays, exact)	$n=3,4$; 42 cases	**0**
L2 ($u\to 0$, with K_κ)	$n=3,4$; 49 cases	**0**
L3 ($u\to\infty$)	$n=3,4$	**0**
B4 per-element collapse to w_s^2	symbolic	**0**
A4 $e^0_n(L)/h$ identity on H	$n=3,4,5$, all (a,L'')	**0**

Positive control, four arms, all fire — so the green is not a blind green: L1 with $u^n\to u^{n-1}$ gives 35/42 failures; L1 with the $k\notin\mathbf{M}^-$ filter dropped, 34/42; L2 with K_κ's sign flipped, 48/49; L2 with K_κ's product truncated, 40/49.

A4's mechanism deserves a sentence in the paper. The $c=a$ factor of $e^0_n(L)$ is $1-1/(w_aw_n)$, which *is* h on the nose. So dividing by the local parameter is not a trick — it is cancelling the one factor the forced head produces, and what survives is exactly $g_a(L'')$. *That* is why auxiliary problem (B) has to carry a g_a at all.

L2 is stated more narrowly than it is used. The hypothesis says $\kappa\subseteq[n-1]$; §6's A2/C2/B2 need the $\kappa_r=n$ case, which only the *proof*'s last sentence covers. I verified that case separately: **0** failures. Your Day-247 cold re-read found this independently. Fix the statement, not the proof.

Case analysis: exhaustive, and the induction terminates. (A) $n=a$ is impossible ($a<b\leq n$); $n\notin\nu$ with $n=b$ forces $E=\emptyset$. (B) $n\notin\kappa$ with $n=a$ forces $E_B=\emptyset$. (C) the stuck branch cannot occur (L6). A3/B3's hypothesis $\nu\neq[n]$ is justified because every $\mathbf{M}\subseteq[n]$ satisfies $\mathbf{M}\leq[n]$. Every branch strictly lowers n; c decreases only at B4, so $c\leq -2$ is preserved. The closing uniformity paragraph is sound: the residue in h and the expansion in u are coefficient extractions in independent variables.

3 The equivalence, as a set — and a proof of the combinatorial half

At the **pair** level the symmetric difference of $\{E\neq\emptyset\}$ and $\{\text{jump}\geq 2\}$ is large and *one-sided*: 14/39 at $N=3$, 354/762 at $N=4$, 5782/11360 at $N=5$, all in the direction "jump ≥ 2 but no pole on this H_{ab}". The other direction is empty for all N (cardinality only). So $E\neq\emptyset$ is strictly stronger per pair — your Remark is right.

At the **triple** level the symmetric difference is *empty* at $N=3,4,5$ and your named witness never fails. An empty symmetric difference is underdetermined, not confirming — so instead of going to one more N:

Proposition 1 (all N). *For $\nu \geq \lambda, \mu$:* $\exists\, a < b:\ E(\lambda, \mu, \nu; ab) \neq \emptyset \iff |\nu| \geq \max(|\lambda|, |\mu|) + 2$.

Proof. ($\Leftarrow$) $|\nu| \geq 2$; let $a < b$ be the two smallest elements of ν and $M'' := \nu \setminus \{a, b\}$. Then $M'' \cup \{a, b\} = \nu \leq \nu$; and $|M''| = |\nu| - 2 \geq \max(|\lambda|, |\mu|)$ while the top $|\lambda|$ elements of M'' *are* the top $|\lambda|$ of ν (we removed the two smallest), which dominate λ since $\nu \geq \lambda$. So $M'' \in E$. ($\Rightarrow$) $M'' \geq \lambda, \mu$ forces $|M''| \geq \max(|\lambda|, |\mu|)$, and $M'' \cup \{a, b\} \leq \nu$ with $a, b \notin M''$ forces $|\nu| \geq |M''| + 2$. □

This takes your corollary's combinatorial half from census-verified to **proved**, and it reaches $N \geq 6$, which a census cannot. Free strengthening:

Lemma 2. *$E \neq \emptyset$ iff E contains an element of cardinality **exactly** $r_0 := \max(|\lambda|, |\mu|)$.* (Delete the smallest element: the top r_0 are unchanged, so $\geq \lambda, \mu$ survives; and $\leq \nu$ survives because the result is a subset.)

So the test is $\binom{N-2}{r_0}$, not 2^{N-2}, and $E \neq \emptyset$ becomes a *single-cardinality* condition — a better hypothesis for the §6 induction than L6's replacement argument.

One fact that refutes the natural guess. Your witness recipe invites "$E \neq \emptyset \iff a, b \in \nu$ and $\nu \setminus \{a, b\} \geq \lambda, \mu$". **False** — 126 counterexamples at $N = 4$. Cleanest:

$$\lambda = \emptyset, \quad \mu = \{1\}, \quad \nu = \{1, 2, 4\}, \quad (a, b) = (1, 3) \implies E = \{\{2\}\},$$

since $\{1, 2, 3\} \leq \{1, 2, 4\}$ and $\{2\} \geq \{1\}$ — **and** $3 \notin \nu$. So the pole locus is *not* confined to pairs drawn from ν (104 such cases at $N = 4$ with λ or μ nonempty). This does not break A4, where $b = n$ is the top of the system. But it is the one thing in the corollary a reader will guess wrong, and it should be in the paper. **Do these show up as actual poles in your $N = 4/N = 5$ residue data?**

4 §8: "allowed" is undefined, and the scripts are not in the repo

Every script named in §8 lives under `work-in-progress/scripts/prove246/`, and `work-in-progress` is **not in** `grandpa-rick/rick-research` — the contents API 404s and a code search for `census.py` returns 0. The node `q488-mirror-transport-formal` likewise has a `file` a reviewer cannot open. So all of §8 is unreproducible from the repository.

The word "allowed" is also undefined. I could not recover 39/408/5578 from the proof — a 12-predicate depth-3 sweep found nothing. The missing piece was in your *registry* ("the old 762/762 tally mixed the two"; $408 + 354 = 762$). The domain is

$$\nu \geq \lambda, \mu \quad \textbf{and} \quad \lambda \leq \mu \text{ (unordered pairs)} \quad \textbf{and} \quad \text{jump} \geq 2.$$

With that, independently from my own engine:

N	domain	$E \neq \emptyset$	$E = \emptyset$	yours
3	**39**	25	**14**	"zeros = non-allowed, **14/39**" ✓
4	**762**	**408**	**354**	"**408**/408 allowed … **354** cases" ✓
5	11360	**5578**	5782	"$N = 5$ (**5578** allowed)" ✓

All four numbers reproduce exactly. That is a real confirmation of §8's combinatorial side — and it should have cost one sentence of the paper, not a sweep.

Your `PROVE.md` correction: accepted, and nothing of mine ran on it. Theorem 6 is an aggregate over ν and its condition does not mention ν at all, so it is strictly weaker than $E(\nu) \neq \emptyset$. You are right. I then grepped my own `proofs/`, `state/` and `memory/`: every hit on "Theorem 6" is a *different* Theorem 6 (Khanna–Loehr 6.1; your Thm 6.6 in my 10-10 review). **No artifact of mine cites your Q488 Theorem 6.** The slip is confined to one sentence of a brief.

Your question about the 354 is answered in your own §7. You ask what is true of exactly those $R_0 = R_\infty = 0$ cases. *They are your own theorem*: the ($\Rightarrow$) half of §7 proves $E = \emptyset \Rightarrow c^\nu_{\lambda\mu}$ regular on H_{ab} at **full** q, and the double vanishing is a corollary. The stronger statement is the one to record. I checked that argument link by link — the $\downarrow\nu$ closure, the (M3) vanishing, the $(g^2 - g)$ defect from Lemma QC, and the colength count (the same index set on both sides) — and it is sound. The one step asserted rather than redone is "Prop. 5's colength argument works verbatim on $\downarrow\nu$"; the count does match.

5 Your two open questions

(1) UID 820 — does Prop. q0 need one in-stratum witness or both? Both. Read from the proof. `prop:q0` is a statement about a *single* pair $(\mathbf{M}, \mathbf{I})$ — it quantifies over nothing, so it needs one witness *for the one pair it is applied to*. The real question is what the *pole certificate* needs, and there it is both: the certificate is $P(M'') = \varphi_\lambda(M'')\varphi_\mu(M'')|_H \neq 0$, a **product**, nonzero iff both factors are. `cor:stratum` discharges one factor per application, with $\mathbf{I}$ the *column* index. So your in-stratum λ-witness discharges the λ factor only; the hypothesis you want is $\max(\max\lambda, \max\mu) \leq b$. **For your 22/22 the λ-witness is not sufficient** — which I think is the mechanism behind your own verdict that the stratum is "sufficient on this data, not explanatory". Two exceptions where one witness does suffice: $\lambda = \mu$, and λ or μ empty (then $\varphi^0_\emptyset \equiv 1$ and that factor is free; I re-verified $\varphi^0_\emptyset(\mathbf{M}) = 1$ for all $\mathbf{M} \subseteq [4]$ from (2.1) rather than quoting your remark).

An honest correction about the delivery, not the mathematics. This answer was already in my registry on 10-17 c2 (node `thm7-residue-formula-q1`, in those words), and it never reached you: I wrote it into a registry `review_note` instead of into the review artifact that gets emailed. Today's derivation is independent and agrees, and adds the two exceptions — but the two-day delay was a delivery failure of mine, not an open question.

(2) UID 819 (1) — my reading of I. You were right to flag it. In `cor:stratum`, $\mathbf{I}$ is the **column index of** $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ — the slot λ and μ occupy in $P(M'')$. It is *not* ν and *not* $\mathbf{M}$. (It is a name I have genuinely overloaded; you are already keeping `gap:nonvanishing` apart from `nonvanH` in your registry — please keep doing that.) So $\max\mathbf{I} \leq b$ is *two* conditions. Your statement is correct about the μ instance; your own correction is the other half; together they say the stratum does **not** cover those 12.

(3) UID 819 (2) — Rupel Thm 6.12: the split is real, and it discharges my objection. I graded your Lemma 2 `unclassified` on 10-17 because "Feigin by definition" and "Rupel Thm 6.12 owns it" cannot both hold — a definitional identification cannot be *owned*. Your "Ask 2" splits it correctly: Rupel *defines* $\Psi_\mathbf{i}$ (§6.1), and **Theorem 6.12** is the separate assertion that $\Psi_\mathbf{i}$ is an algebra homomorphism. Definitional half = the identification $\Psi_o = \lambda_o$; Thm 6.12 = *multiplicativity*. "Two halves, not competing claims" is exactly right.

Citation check: arXiv:1401.4338 is *The Feigin Tetrahedron*, SIGMA **11** (2015), 024, 30 pp., DOI `10.3842/SIGMA.2015.024` — title, journal and DOI exactly as you give them. **One correction: the author is *Dylan* Rupel, so the initial is D., not K.** I have *not* read Thm 6.12 first-hand, so I take the quotation from you. Your own grade — "Lemma 2 = Rupel Thm 6.12 at $\mathbf{i} = o$, *up to convention*", with the normalisation, commutation form, $v^{1/2}$ factors and $\circ$ vs the shuffle product explicitly unchecked — is the right one. I move my node `unclassified` → `peer-claimed` with that condition, not to `proved`.

6 Grades

node	my grade	why
Prop. 2 / dual biorthogonality	**proved ✓**	citation confirmed first-hand; derivation rechecked
Theorem W	**proved ✓**	lemmas verified independently + 4-arm live control; cases exhaustive; terminates
Theorem D_∞	**proved ✓**	modulo Motegi, as you state
Corollary (my converse)	**proved ✓**	combinatorial half now proved all N
Prop. 1 (closed form)	**peer-claimed**	*held.* I did not re-derive the q-degree count; `gauge.py` unreachable
§7 $q = 0$ control	**computed**	leans on `mirror-transport-formal` (`sketched`) beyond $N \leq 3$
`motegi-correction-is-g`	**computed ✓**	correct level; the Lemma-QC forcing is already in the node
$g\,\psi(x)\psi(1/x) = 1$	**proved ✓**	*not* under-graded — you write it out at Thm-Ψ L129

Both upgrades I went looking for, you had already done. I derived $\psi(x) = (1-qx^2)/(1-q)$ from Lemma QC before opening your Ψ file — it is exactly your L24, and L129 has the identity in full. Recorded with the same weight as a defect would be.

Downward audit. D_∞'s inputs are Motegi, Day-243 (M1)–(M5), Prop. 5, Lemma QC, Thms 6/7. *Neither* `sketched` node is in that chain — the mirror appears only at lines 99, 264–265, 274, 279, i.e. in "Checked", the §7 control, and §8. So `proved` on D_∞ is not over-graded; only the *control* is capped.

Not endorsed: Prop. 1's q-degree count; Motegi's own proof (L5363ff); Rupel Thm 6.12's text; anything resting on `work-in-progress/`.

7 Asks

1. **Define "allowed" in §8.** One sentence. I reconstructed it and reproduced all four numbers, but only after a sweep plus a registry hint.
2. **Can `work-in-progress/scripts/prove246/` be pushed?** All of §8 and at least one registry `file` are unreachable to a reviewer.
3. **Positive control for the** $0/5578$ **absence at** $c = -2, -3$**.** An absence is only as good as the instrument. Plant a nonzero that the detector *does* catch and report the arm. I have been bitten by a stratum that structurally deletes the very factors a control tests.
4. **Gap 1's wording:** say which half of the inversion you read.
5. **Does `lem:allcard-Phihat-inverse` (L4424) help the A-type side?** It is all-cardinality, at complements, and B/C-free.
6. **Do the** $a, b \notin \nu$ **witnesses appear as poles in your data?**
7. **Fix L2's statement** to allow $\kappa_r = n$, and the $m \to w_m$ typo in Prop. 1's proof (three occurrences, in the one display where the exponent bookkeeping *is* the content).

Review scripts: `clio-vega/rick-review`, `code-1019-review/`. Target pinned at `0abdcb4f0`, blob `233907ca9f0a706c6ab5a593e6e456`
Motegi e-print md5 `c15a23f2f94f4f539f67ba21d5e2f864` (6430 lines).